Zitao Chen

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PROFILE

I am an undergraduate student in the Department of Computer Science and Technology at Tsinghua University. My research focuses on AI for Science and LLM agents, with an emphasis on geometric generative modeling, molecular discovery, and self-improving agent systems. My research interests include:

- **Geometric Molecular Generative Modeling:** Developing E(3)-equivariant 3D generative models and fixed-dimensional representations for molecular discovery and de novo drug design.
- **Molecular Identification & Statistical Thermodynamics:** Applying machine learning to MSn-based molecular fingerprint identification, Boltzmann generation, and free-energy estimation.
- **Self-Evolving Agent Systems & Evaluation:** Designing autonomous harnesses that improve through execution-trace analysis and rigorously evaluating AI systems that build better AI.

EDUCATION BACKGROUND

Tsinghua University Aug 2023 - Jul 2027
Department of Computer Science and Technology, Undergraduate Student

- **GPA**: 3.9/4.0, Rank: 8/172
- **Awards & Honors**: Tsinghua Spark Program (星火计划, 19th Cohort, 2025–Present); Beijing Natural Science Foundation Qi Yan Grant (北自然"启研"计划, 2025–2026); Tsinghua Academic Promotion Program (学推计划, A-Level Project, Project Lead, 2025, 2026); Tsinghua Comprehensive Excellence Scholarship (综合优秀奖学金, 2024, 2025); Jittor AI Challenge (计图AI挑战赛, Second Prize, Human Skeleton Rigging Track, 2025); Tsinghua Challenge Cup (挑战杯, Second Prize, University-Level Competition, 2025); Mathematical Contest in Modeling (MCM/ICM, Meritorious Award, 2025); Tsinghua AI Challenge (人工智能挑战赛, First Prize, 2024); Tsinghua Quadruped Robot Development Competition (机器狗开发大赛, First Prize, 2024).

PUBLICATIONS

- **Zitao Chen**, Jiatong Ji, Shuming Liu, & Jian Ma. *SEAL-BG: Scaling Boltzmann Generation with Equivariant Perception and Local-Coordinate Densities*. Under review.
- Alex Zhou, **Zitao Chen**, Guan Wang, Yinjun Jia, & Wei-Ying Ma. *MolGAN: Few-Step 3D Molecule Generation with Hierarchical Reasoning*. Under review.
- Bohan Lyu, Yucheng Yang, Siqiao Huang, Jiaru Zhang, Qixin Xu, Xinghan Li, Xinyang Han, Yicheng Zhang, Huaqing Zhang, Runhan Huang, Kaicheng Yang, **Zitao Chen**, Wentao Guo, Junlin Yang, Xinyue Ai, Wenhao Chai, Yadi Cao, Ziran Yang, Kun Wang, Dapeng Jiang, Huan-ang Gao, Shange Tang, Chengshuai Shi, Simon S. Du, Max Simchowitz, Jiantao Jiao, Dawn Song, & Chi Jin (2026). *MLS-Bench: A Holistic and Rigorous Assessment of AI Systems on Building Better AI*. In *Advances in Neural Information Processing Systems* (NeurIPS 2026).
- **Zitao Chen**, Jiatong Ji, Yinjun Jia, Wei-Ying Ma, & Yanyan Lan[†] (2026). *MolAlign3D: Enhancing Fixed-Dimensional E(3)-Equivariant Latent Space for High-Fidelity 3D Molecular Reconstruction and Editing*. In *Forty-third International Conference on Machine Learning* (ICML 2026).
- **Zitao Chen**[*], Yinjun Jia[*†], Zitong Tian[*], Wei-Ying Ma, & Yanyan Lan[†] (2025). *Manipulating 3D Molecules in a Fixed-Dimensional E(3)-Equivariant Latent Space*. In *Proceedings of the 39th Conference on Neural Information Processing Systems* (NeurIPS 2025).

RESEARCH EXPERIENCES

MSn-Based Small-Molecule Fingerprint Identification **Apr 2026 - Present**
Research Intern, Coley Lab, Massachusetts Institute of Technology

- Developing machine learning methods for small-molecule fingerprint identification from multi-stage tandem mass spectrometry (MSn) data.
- Leveraging MSn fragmentation patterns to support molecular structure elucidation and compound identification in metabolomics and drug discovery pipelines.

Tractable-Likelihood Generative Modeling & Free Energy Calculation **Jul 2026 - Sep 2026**
Visiting Student, Ma Lab, Carnegie Mellon University

- Developing generative models with tractable likelihoods to accurately model Boltzmann distributions for free energy calculations.
- Developed SEAL-BG, combining SE(3)-equivariant geometric perception with local-coordinate densities for scalable Boltzmann generation and free-energy estimation.

Self-Evolving Harness Research **May 2026 - Jul 2026**
Research Intern, Zhipu AI

- Proposed U-Self, a self-evolving agent harness where the same agent reads its own execution traces, diagnoses failures, and iteratively evolves its prompt, skills, and memory without an external debugger or human intervention.
- Designed a seven-stage self-contained evolution loop with strict data isolation across calibration, holdout, and final sets to promote generalization rather than overfitting to a fixed dataset.

Unified Semantic-Generative 3D Molecular Modeling **Sep 2025 - Feb 2026**
Research Intern, AIR, Tsinghua University

- Proposed MolAlign3D, a unified framework that anchors fixed-dimensional E(3)-equivariant latent spaces with pretrained molecular semantics, jointly supporting high-fidelity 3D reconstruction, property prediction, and zero-shot molecular editing.
- Implemented the full pipeline and conducted all experiments. Drafted the majority of the manuscript.

Hierarchical Reasoning Model for Drug Discovery **Aug 2025 - Sep 2025**
Research Intern, Sapient Intelligence

- Contributed to applying Hierarchical Reasoning Model to AIDD tasks, including binding affinity prediction and Structure-Based Drug Design.

3D Fixed-Dimensional Molecular Autoencoder **Mar 2025 - May 2025**
Research Intern, AIR, Tsinghua University

- Developed a 3D molecular autoencoder with an E(3)-equivariant fixed-dimensional latent space, enabling molecular sampling, interpolation, and flexible variable-atom editing.
- Served as a core contributor, leading data curation, implementing key model modules, and contributing substantially to manuscript preparation.

EXTRACURRICULAR ACTIVITIES

Vice President Jul 2026 - Present
Student Association for Science & Technology, Tsinghua University

- Organize university-wide science and technology programs, including the Science Camp and Liaoyuan Forum.

Vice President Jun 2025 - Jun 2026
Student Association for Science & Technology, Department of Computer Science and Technology, Tsinghua University

- Led the Network Department, coordinating cross-departmental events, technical workshops, and external communications.

Deputy Minister, Learning & Development Feb 2025 - Jun 2026
Tsinghua University Student Union

- Organized and managed campus-wide learning programs and peer-support initiatives to promote academic engagement and student development.
- Coordinated and hosted academic sharing sessions led by senior students, providing guidance on coursework, research, and personal development.

Core Member Jun 2024 - Present

Student Union, Department of Computer Science and Technology, Tsinghua University

· Support organizational operations, outreach activities, and liaison work between student groups and faculty members.

· Contributed to successful coordination of departmental events, improving student participation and satisfaction.

SKILLS AND INTERESTS

- **Language Skills**: Chinese (native); English (proficient)
- **Technical Skills**: Machine Learning, Python, Software Development, PyTorch
- **Interests**: Badminton, Photography, Guitar

DECLARATION

I hereby declare that all the details furnished above are true to the best of my knowledge and belief.